

IMPERIAL

Carbon Accounting and Reporting

Lecture 8 – Carbon Credits ISO Net Zero Standard

Dr James Robey / Jon Hampson
7th November 2025

Course Overview

COURSE OVERVIEW

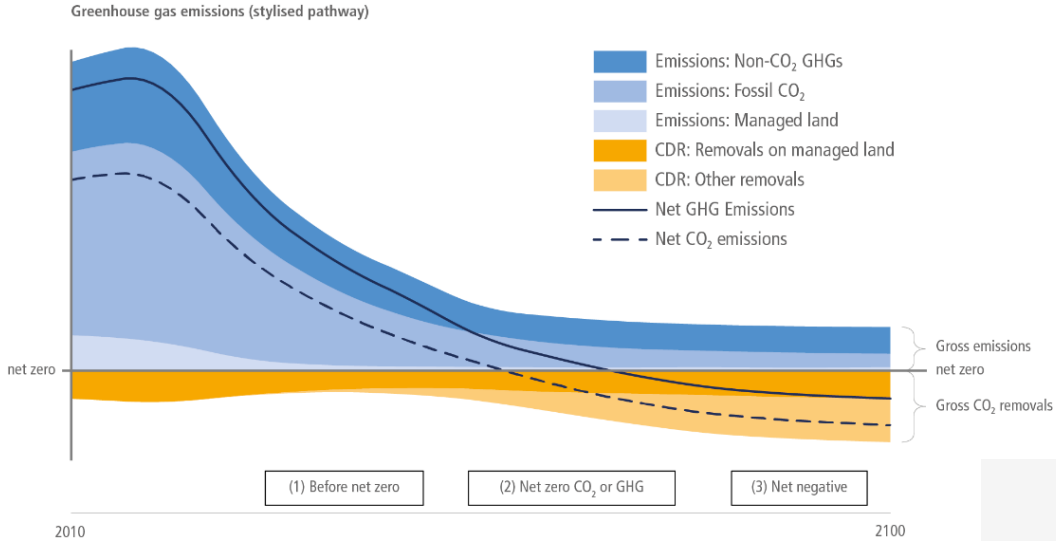
- I. Sustainability – the Big Picture
- II. Carbon and Climate Change
- III. Basics of Carbon Accounting
- IV. Introduction to Organisational Carbon Accounting
- V. Scope 2 – energy emissions
- VI. Scope 3 – indirect emissions
- VII. Net Zero, Science Based Targets and Carbon Credits
- VIII. Sustainability reporting & materiality

Carbon Credits

Stuart Leckie

Head of Carbon Credit Programme, Capgemini

The challenge



Source: IPCC

The bathtub analogy

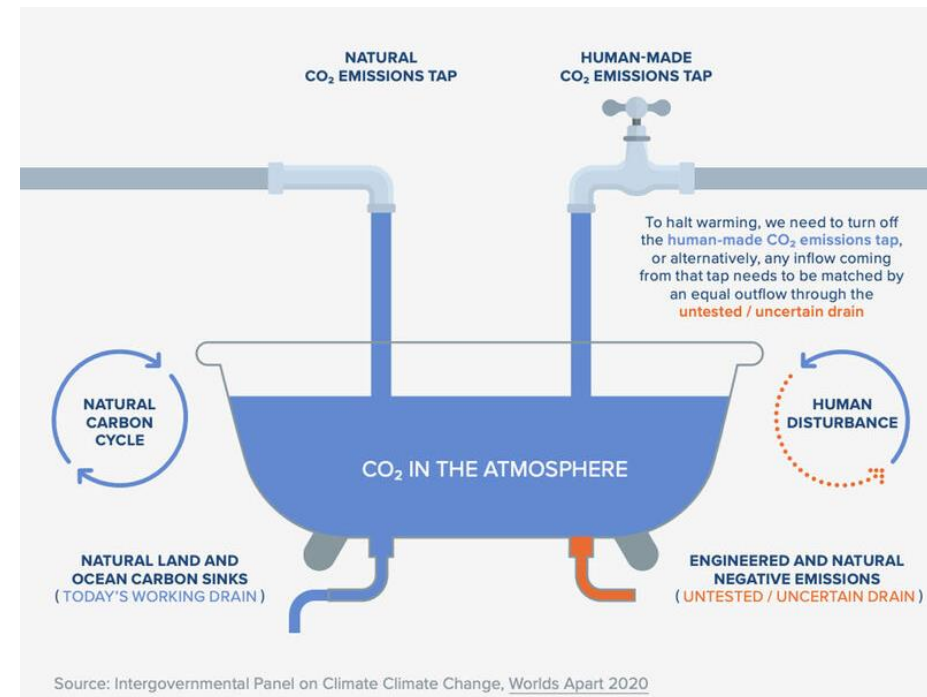
To stop your bath overflowing you can reduce the water from the taps, but - unless you turn them off completely - the water level will still rise.

The only way to balance the level while they're still running is to pull the plug and drain the same amount that that taps are putting in. When that happens, the bath reaches the equivalent of net zero.

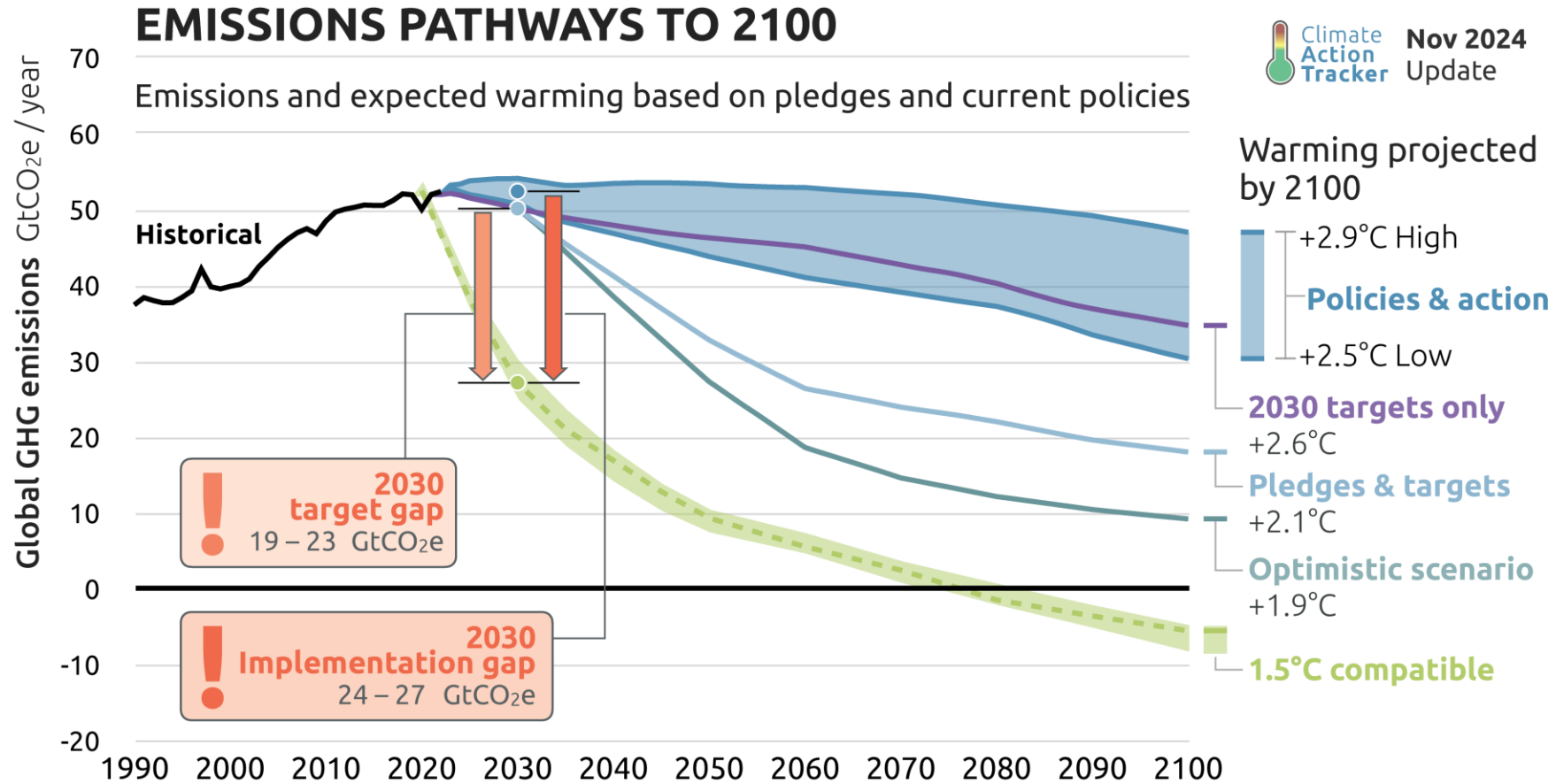
To avoid the worst effects of climate change we need:

- dramatic emission reductions
- increased removals from the atmosphere

The more rapidly this occurs the lower the potential impacts.



Where we are today



What can companies do to close the target and implementation gaps?

- Reduce their emissions and collaborate with their value chain
- Support stronger policies
- Take voluntary actions beyond their own value chain

First, a definition of a carbon credit and carbon offsetting



A **carbon credit** is a tonne of carbon dioxide equivalent that has been avoided entering/removed from the atmosphere.

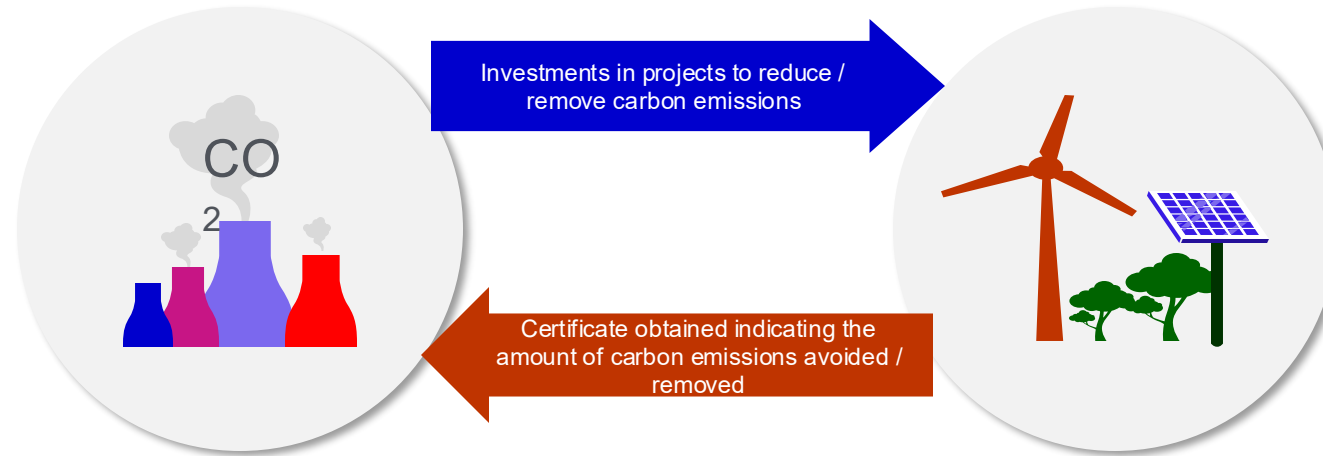
Carbon offsetting means buying carbon credits equivalent to your carbon impact. This means you compensate for every tonne of CO₂ you emit by ensuring there is one tonne less in the atmosphere. And because one unit of CO₂ has the same climate impact wherever it is emitted, the benefit is the same wherever it is reduced or avoided too.



What does carbon offsetting mean ?

KEY TAKEAWAYS :

1. The underlying principle of carbon offsetting is that the impacts of one tonne of carbon emitted somewhere can be offset by sequestering or reducing another tonne of carbon elsewhere
2. Carbon credits don't always have the exact characteristics equivalent to the emissions made which makes direct offsetting challenging



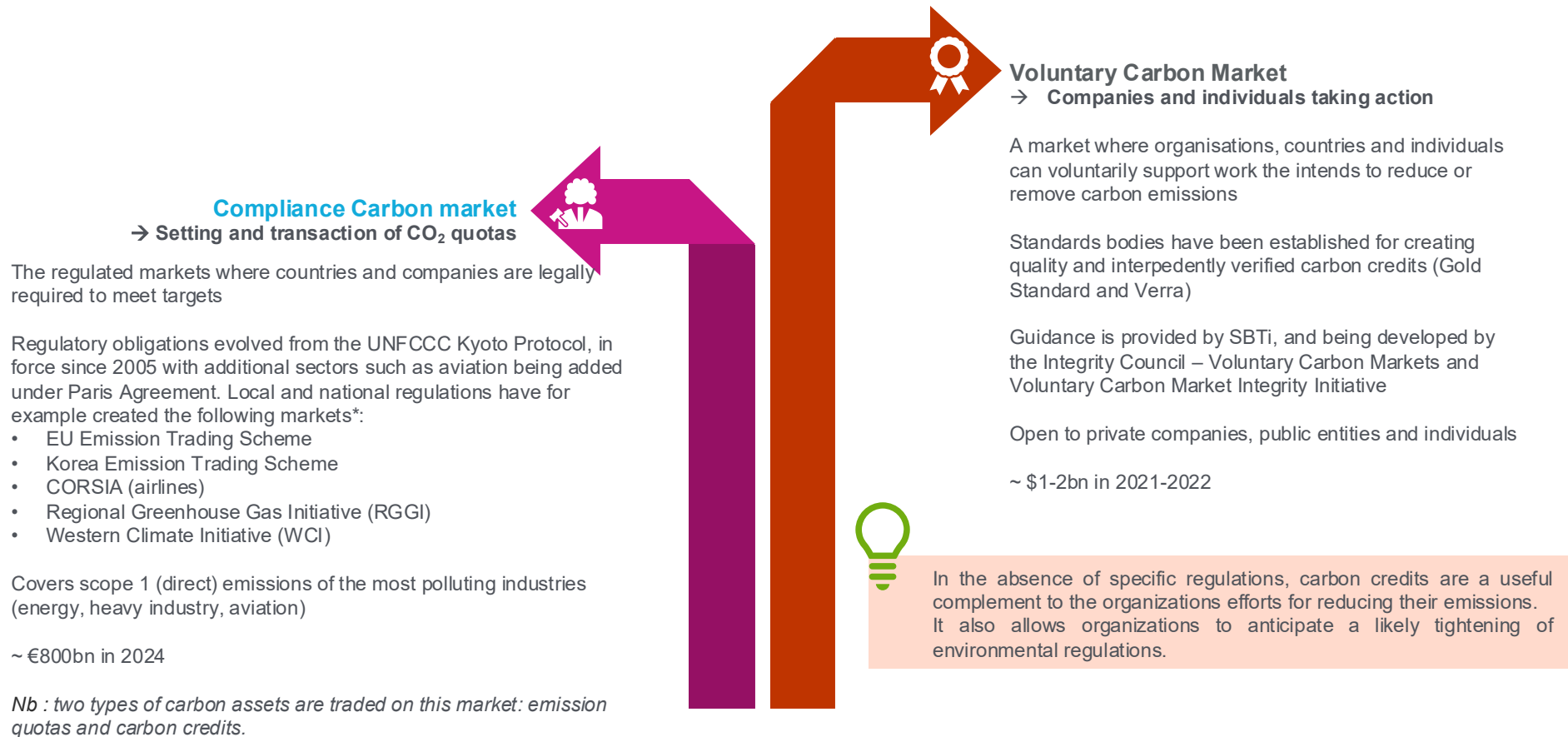
- Companies and governments need to meet their emissions targets.
- Even when they do a lot to reduce their carbon footprint, most of them can't reduce it to zero

- The investments goes to carbon credits from various different types of projects that reduce / remove CO₂ emissions

There are two types of carbon markets to tackle climate change

KEY TAKEAWAYS :

1. The voluntary carbon market is much smaller than the compliance carbon market
2. Carbon offsetting is often criticized because associated with greenwashing. However, carbon credits constitute an interesting lever when it complements an ambitious emission reduction strategy



Different types of carbon credit projects

KEY TAKEAWAYS :

1. There are two main categories of carbon credits: removal or avoidance. They are differentiated by the way they fight carbon emissions
2. The most in demand carbon credit projects are forestry (afforestation, reforestation)
3. Most carbon credits come from low and middle incomes countries

Avoidance or removal?

Types of carbon credit projects


Avoidance
An avoidance project means another party does not produce Greenhouse Gases that they otherwise would have. This category can be split between avoided emissions and emission reductions.


Removal
A removal credit project takes CO₂ from the atmosphere and stores it.



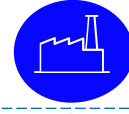
1 Energy Supply

- Grid-connected renewable electricity
- Off-grid renewable electricity



2 Energy Demand

- Improved cookstoves
- Transport modal shifting
- Building systems



3 Industrial Emissions

- Carbon capture and storage
- Methane capture
- Industrial gases



4 Nature Conservation

- Avoided deforestation (REDD+)*
- Jurisdictional REDD+
- Improved agricultural practices

*Reducing Emissions from Deforestation and Degradation (REDD+)



5 Nature Restoration

- Afforestation /Reforestation
- Agroforestry
- Soil Carbon
- Mangroves



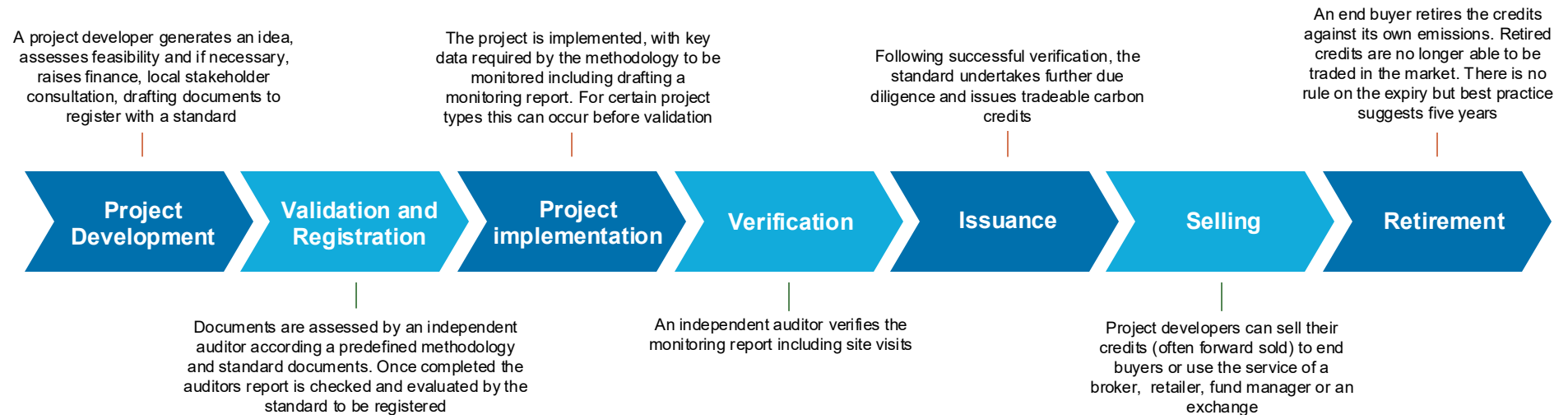
6 Technology / Engineered removal

- Direct Air Capture and Storage (DACs)
- Enhanced weathering
- Biomass Carbon Removal and Storage (BiCRS)

Lifecycle of a carbon credit

KEY TAKEAWAYS :

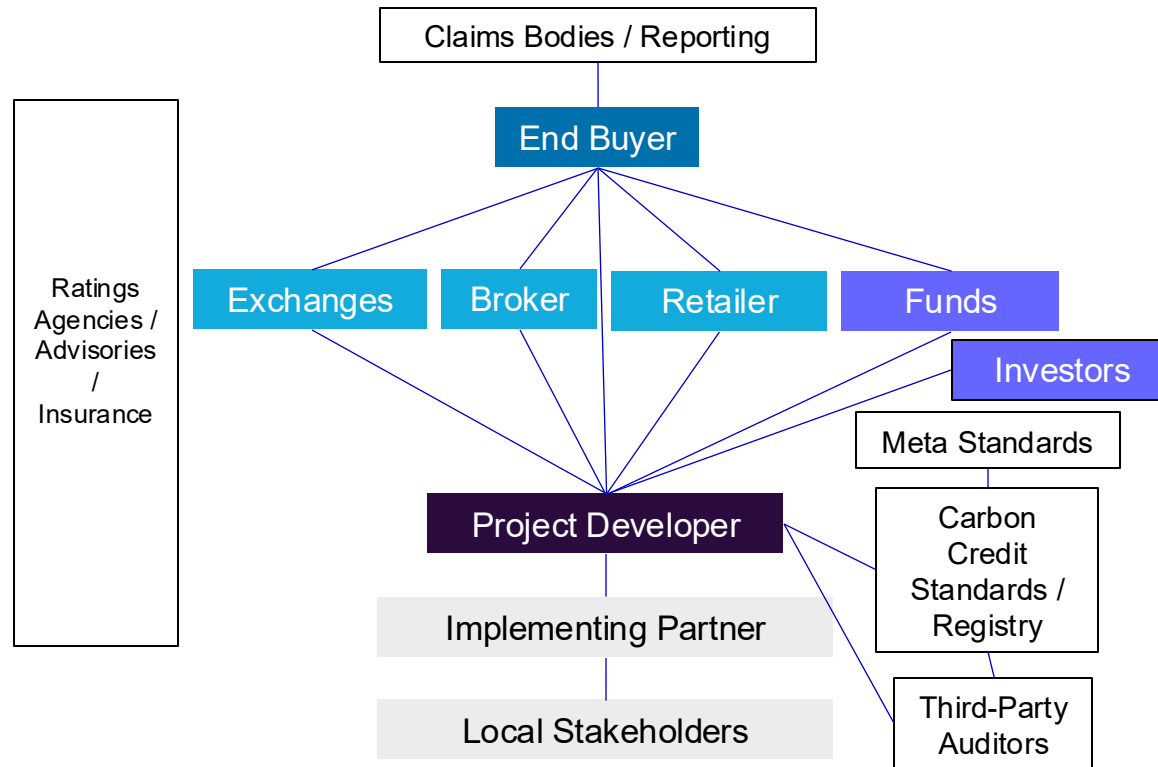
1. Carbon credits represent ex-post emission reductions or removals and go through several steps



Structure of the Voluntary Carbon Market

KEY TAKEAWAYS :

1. Governance structures for the carbon market are still emerging. It is a relatively complex ecosystem for buyers to navigate.

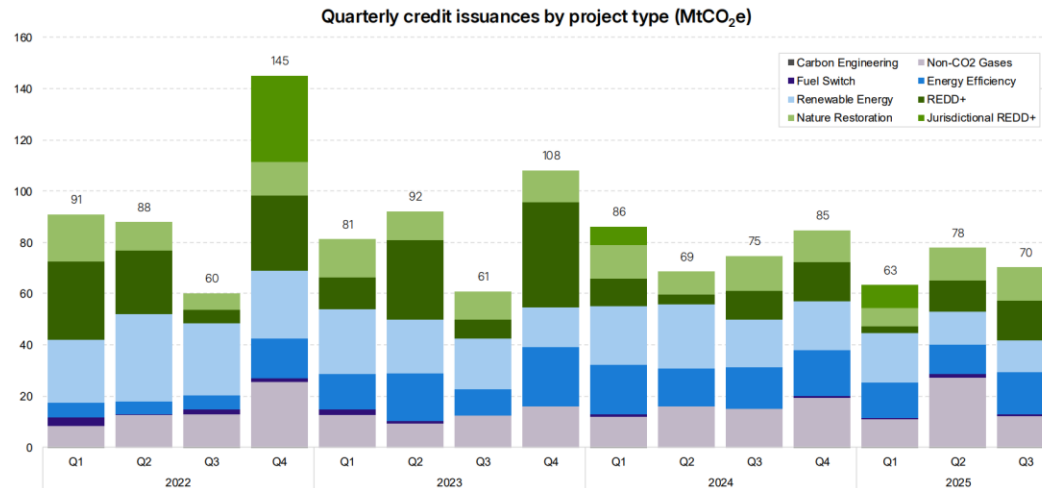


- End buyer: Individual, company or government
- Project developer: Originator of carbon credit project and receives credits after issuance by carbon credit standard after third party verification
- Exchange / Broker / Retailer: Different mechanism by which credits can be procured can be direct from project developer or on the secondary market.
- Funds / Investors: provide capital for the market
- Carbon Credit Standards / Registry: e.g. Verra / Gold Standard that sets methodologies and issues credits to their registries.
- Third party auditors verify carbon credit projects.
- Meta-Standard: e.g. IC-VCM that is attempting to standardize the market
- Implementing Partner: often detailed knowledge of project location to operate the project on a day-to-day basis
- Local Stakeholders: can be both communities and officials that have important part directly or indirectly in the project.
- Ratings Agencies / Advisories / Insurance are establishing themselves to support different actors in the market.
- Claims Bodies / Reporting: provide governance on the reporting and use of carbon credits.
- Organisations can play multiple rolls
- Not all key relationships are indicated on the diagram

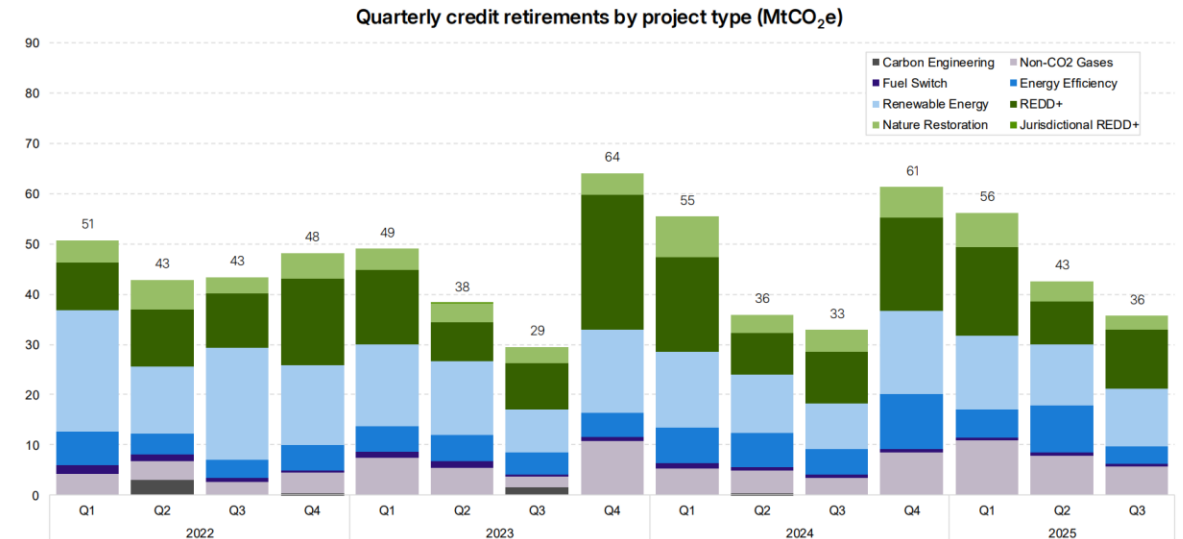
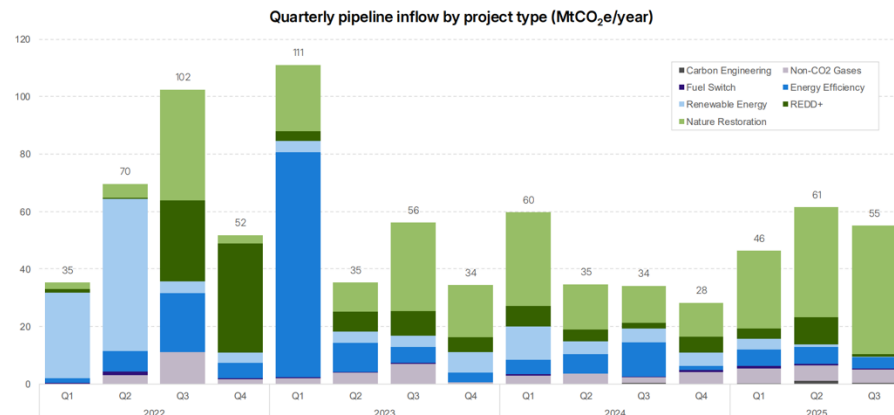
Supply and demand for carbon credits

KEY TAKEAWAYS :

1. The majority of carbon credits are avoidance but removals have an increasing share
2. Issuance is slowing for renewable energy credits
3. Potential size of the market is much greater but faces several barriers



- Demand for carbon credits was growing rapidly through 2021 but has plateaued in the last three years due to integrity concerns and risks of greenwashing.
- Most credits currently available are avoidance with only around 10% of credits are removals with almost all being nature based.
- 315Mt of carbon credits were issued in 2024 but only 185Mt were retired by buyers.
- Companies that have committed to net zero represent several Gt of emissions (1.3Gt with SBTi targets) and therefore much greater potential demand but there is a lack of clear incentives.
- Engineered removals issuance and retirement are currently negligible but long-term purchases are growing. Inflow of nature-based projects is increasing.

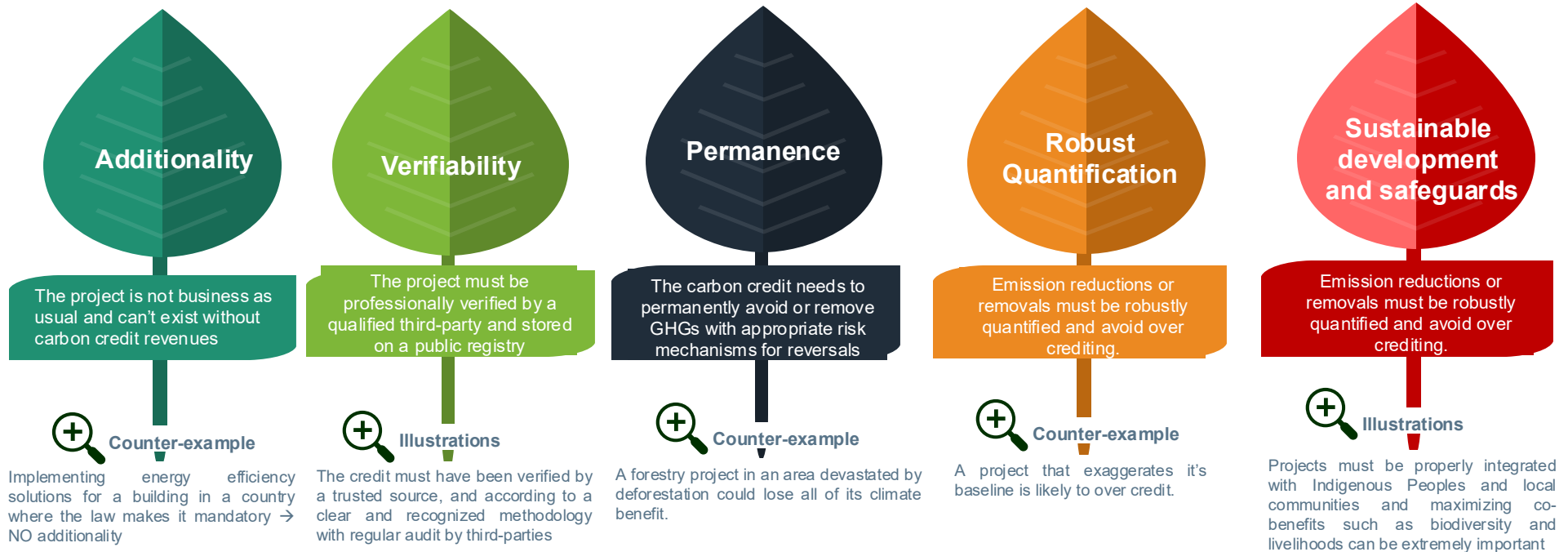


Source: MSCI Carbon Markets

Determining carbon credit quality

KEY TAKEAWAYS :

1. There are several key criteria for a carbon credit project to meet
2. The IC-VCM has published 10 Core Carbon Principles
3. The relative importance of the characteristics that make up quality vary depending on the stakeholder



Carbon credit standards (e.g. Gold Standard and VCS) are intended to ensure the quality of carbon credit credits. The challenge is that quality is not binary due to the multiple criteria involved, critical criteria, like additionality, are a matter of confidence rather than absolute truth. This means that quality exists along a continuum and that due diligence is required all along the chain of purchasing carbon credits.

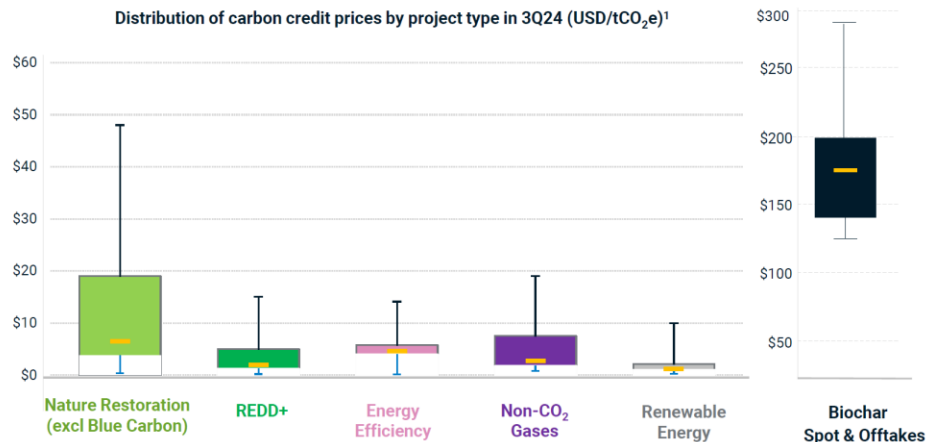
The Integrity Council for the Voluntary Carbon Market Core Carbon Principles

- A. Governance: 1. Effective governance, 2. Tracking, 3. Transparency, 4. Robust independent third-party validation and verification
- B. Emissions Impact: 5. Additionality, 6. Permanence, 7. Robust quantification of emission reductions and removals, 8. No double counting
- C. Sustainable Development: 9. Sustainable development benefits and safeguards, 10. Contribution toward net zero transition

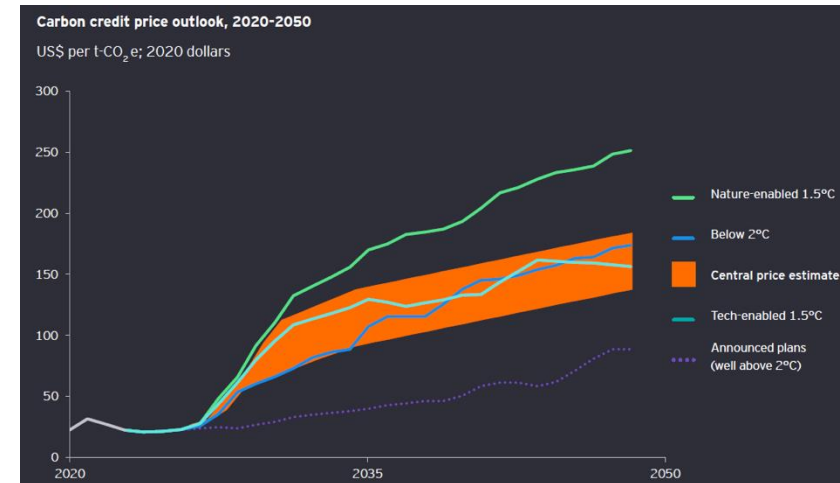
Price of carbon credits

KEY TAKEAWAYS :

1. The Voluntary Carbon Market is not a commoditised market. There are no universally accepted guidelines on how to price carbon credits.
2. Removals projects are generally priced higher than avoidance projects
3. There is large variations in the price of credits



Source: MSCI Carbon Markets



Source: EY, Essential and still evolving: The global voluntary carbon market outlook 2024

The price of a carbon credit depends on many factors



Location

A project based in Western Europe will for example usually be more expensive than one in Asia



Type

Removal projects such as planting trees will usually be more expensive than avoidance projects



Standard

The fee that the project operator had to pay to be certified by a standard will be included in the credit's final price



Co-benefits

A project that provides extra-benefits on top of emission reduction (such as job creation, education...) can command higher prices.

Barriers to the Voluntary Carbon Market reaching its potential

KEY TAKEAWAYS :

1. Concerns over credit integrity and lack of incentives are key barriers to scaling the market.
2. There are several reasons for businesses to act today.

Concerns of carbon credit integrity

- Several concerns have been raised on the integrity of carbon credits issued focusing on over-crediting and involvement and safeguards for local communities.
- The Integrity Council for the Voluntary Carbon Market (IC-VCM) has been established to raise the overall integrity of supply.
- IC-VCM's Core Carbon Principles are beginning to gain support from governments
- Increased willingness to pay is also allowing project developers to improve standards.
- Infrastructure is still developing for the market.

Lack of incentives to invest in carbon credits

- Companies are afraid of greenwashing claims
- There is a lack of clear incentives to retire carbon credits
- The Voluntary Carbon Market Integrity Initiative (VCMI) has launched its Claims Code of Practice and ISO have launched a Carbon Neutral standard (14068).
- SBTi has sent mixed messages on the use carbon credits. Currently use up until the point of Net Zero is 'Beyond Value Chain Mitigation' that is not directly required.
- Many companies struggle to see a clear route to achieving scope 3 reductions without being allowed some use of carbon credits.
- Government standards and incentives are in development.

Why are businesses acting today?

- It's the right thing to do
- Increasing stakeholder expectations
- Preparing for future regulation
- Securing supply ahead of likely shortages
- Seek a competitive advantage
- Growing a trillion dollar industry

Building a portfolio of carbon credit projects is the best way to mitigate risk and take action

In the short-term : (↗)

- Governance mechanisms to determine quality and integrity in the voluntary carbon market will be concluded.
- The demand for carbon credits is expected to increase
- Trends suggest availability will become challenging in the next several years. Supply of credits is inelastic and new projects take years to come to market

In the long-term : (↗)

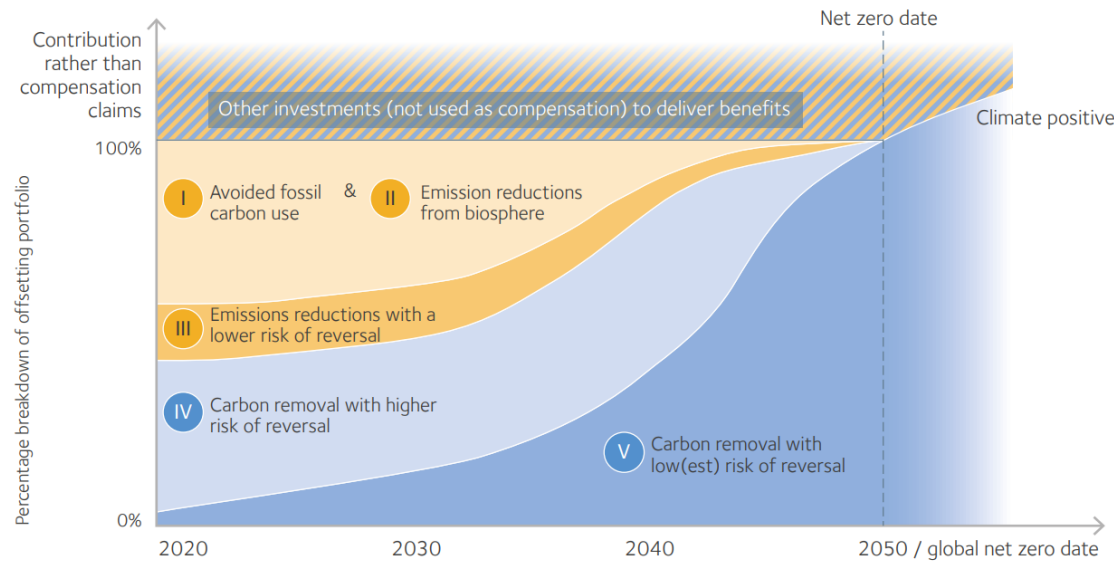
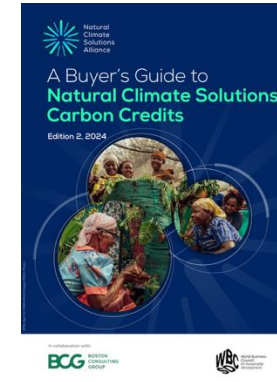
- Removals based credits including novel technological approaches are likely to play an increasing role in the carbon market
- The line between the voluntary and compliance market will become increasingly blurred
- Prices will continue to rise

Building a carbon credit portfolio

KEY TAKEAWAYS :

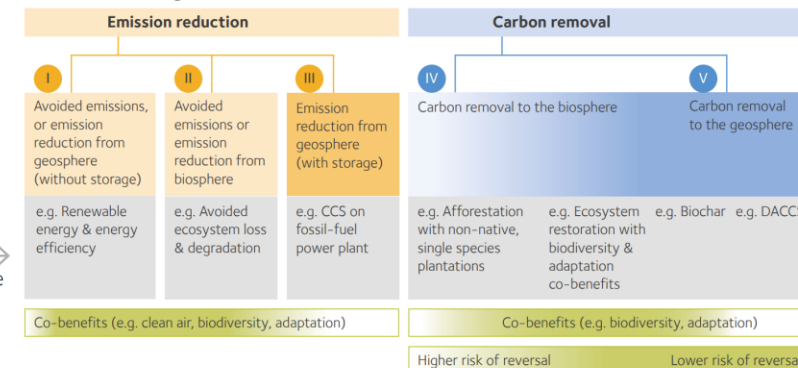
1. Guidance is increasingly available to buyers
2. Best practice is to build a portfolio of different types of credits based on key characteristics

- There is increasing guidance available for buyers on how to transact in the voluntary carbon market
- Climate leaders are increasingly sharing best practices and documents
- Building a portfolio is the best solution to reduce risk and make the biggest impact



Source: Oxford-Principles-for-Net-Zero-Aligned-Carbon-Offsetting-revised-2024

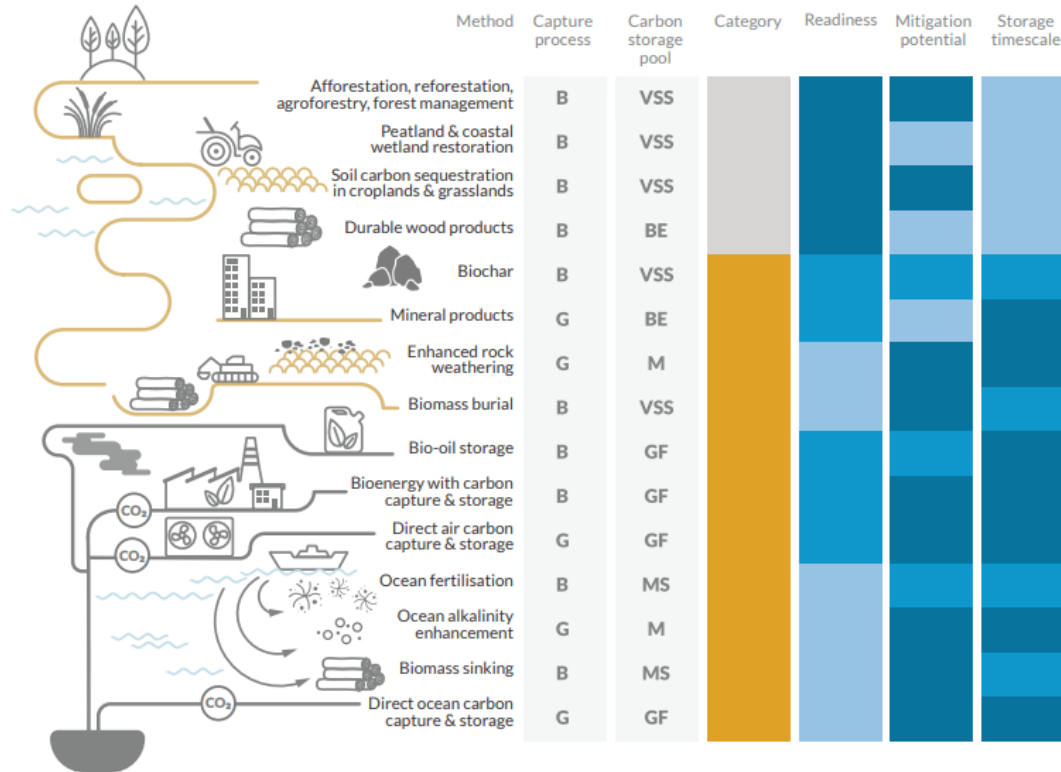
- The Oxford Net Zero Aligned Carbon Offset Principles provide an example of how to build and evolve a carbon credit portfolio
- Starting with a mix of different approaches that transitions over time to removal credits with durable storage.



Focusing on removals in a portfolio

KEY TAKEAWAYS :

1. Due diligence on removals options can be complicated and uncertain
2. Companies should try to maximise their impact
3. Different potential solutions have different characteristics



Legend:

B	Biological	VSS	Vegetation, soils and sediments	Conventional	High	Large	> Ten millennia
G	Geochemical	BE	Built environment	Novel	Medium	Moderate	Centuries to millennia
		GF	Geological formations		Low	Small	Decades to centuries
		MS	Marine sediments				
		M	Minerals				

- Companies need to evaluate different carbon removal options
- Forming alliances can help scale impact
- Acting transparently and sharing best practice is key
- Space needs to be created to allow failure
- Creating deals that are bankable and investable is key.
- Supporting a range of approaches allows the possibility for many routes to achieve the removals we will need for net zero and beyond.

Source: Smith, S. M. et al, The State of Carbon Dioxide Removal 2024 - 2nd Edition.

Questions

1. Who should be accountable if carbon credits are shown to underperform?
2. Should carbon credits be used to count towards net zero targets?
3. What role should companies and the voluntary carbon market play in helping countries achieve their Nationally Determined Contributions under the Paris Agreement?

ISO Net Zero Standard

Shana Gallagher, BSI

ISO Net Zero Standard

British Standards Institution (BSI)
Net Zero Policy Team
November 2025

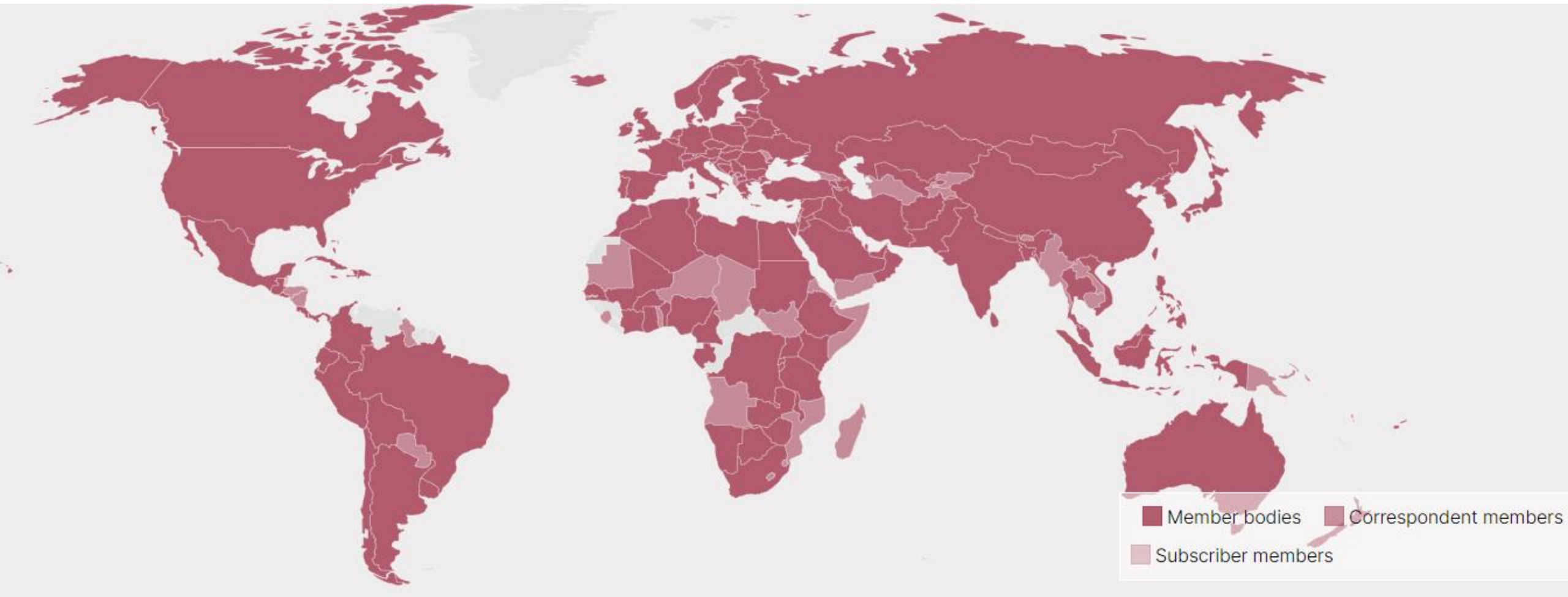
Contact: Shana Gallagher
Email: shana.gallagher@bsigroup.com



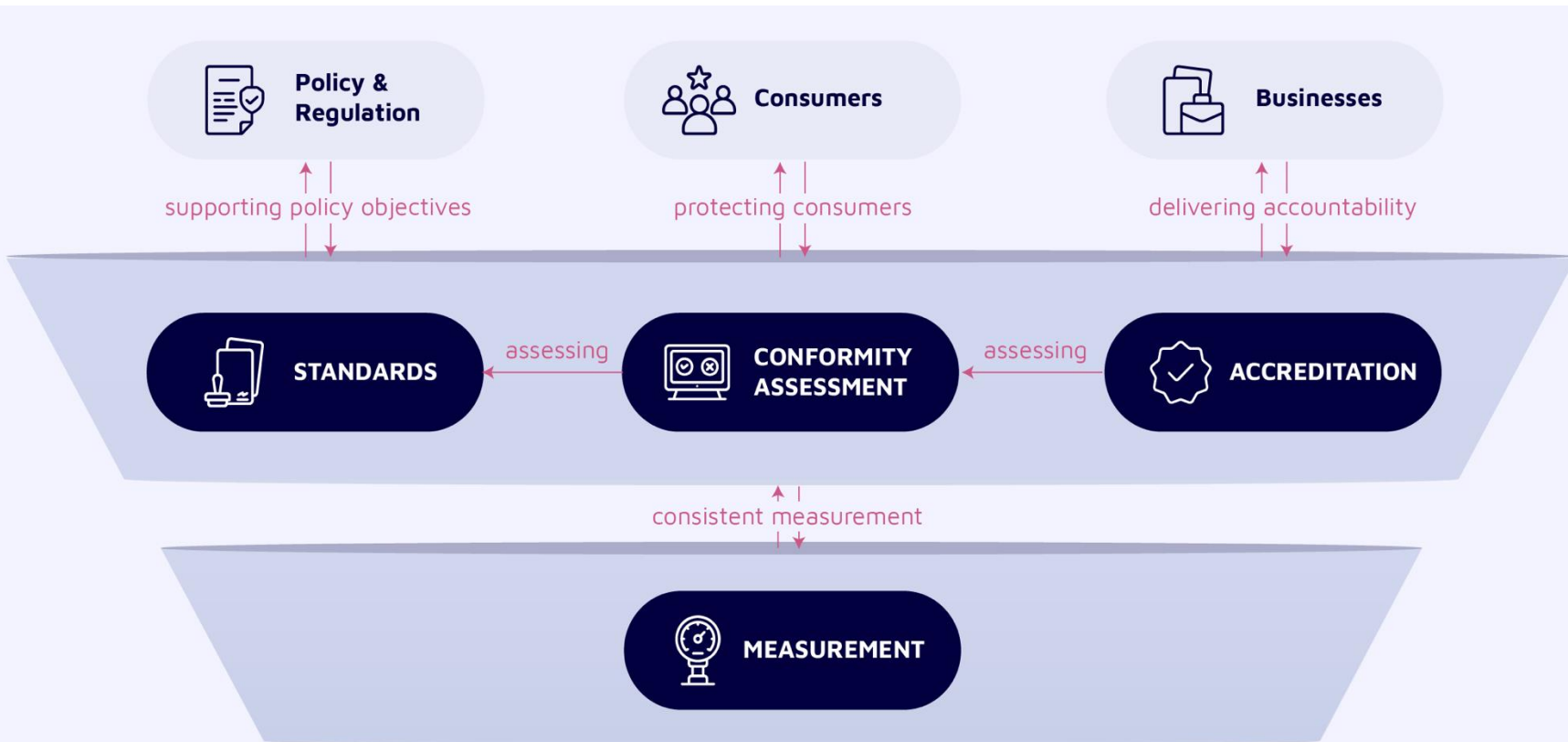
United Nations Climate Change
Global Innovation Hub



International standards that reflect global input



The system that delivers market trust



STANDARDS

Developing consensus on what 'good' looks like. Many countries have a National Standards Body. These national bodies coordinate with each other at international level.



CONFORMITY ASSESSMENT

Assessing that standards are being met across global supply chains.



ACCREDITATION

Ensuring confidence in conformity assessment. Many countries have a National Accreditation Body. These national bodies coordinate with each other at international level.



MEASUREMENT

Ensuring consistent measurements and scientific calibration. Many countries have a National Metrology Body. These national bodies coordinate with each other at internal level.

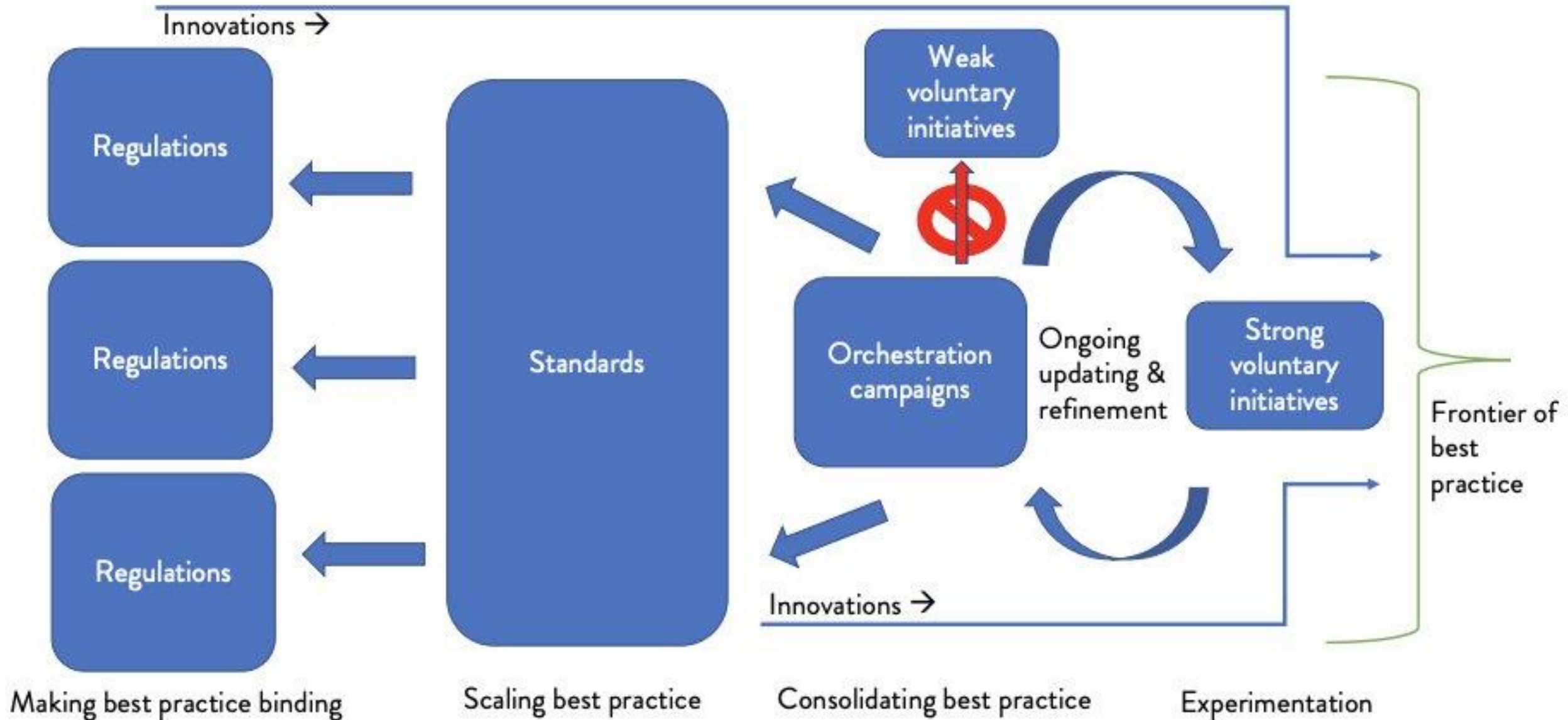


The [“Net Zero Conveyor Belt” Model](#) describes the importance of leveraging Quality Infrastructure to scale and accelerate the net zero transition

Figure 1: A “conveyor belt” governance system for net zero over the next decades

<----- More binding, harder to update, less influence for pro-climate voices ----->

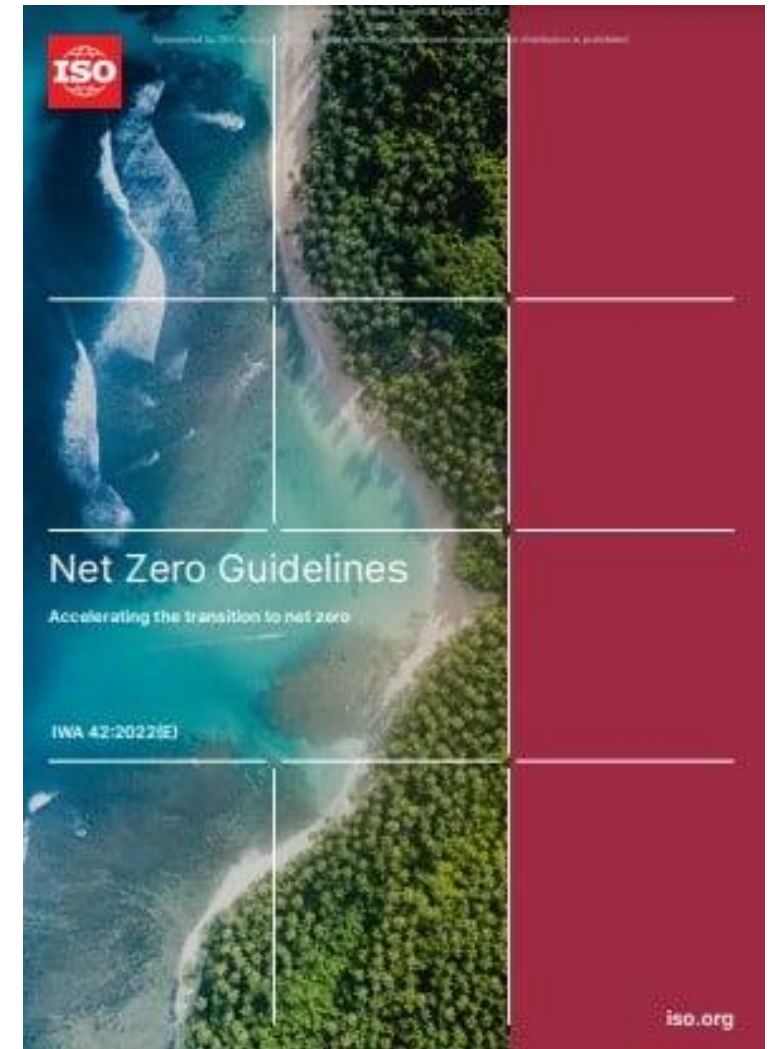
-----Less binding, easier to update, more influence for pro-climate voices ----->



ISO Net Zero Guidelines

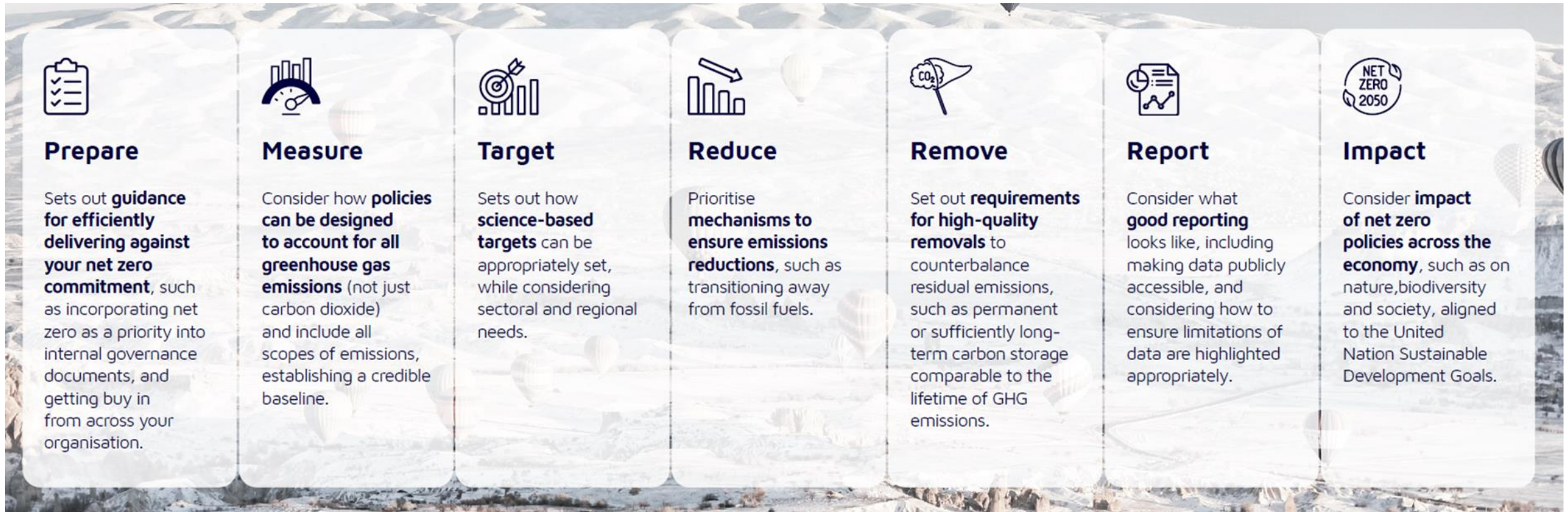
A global basis for net zero action

- Launched at COP27 in 2022 - defined net zero within the ISO system for the very first time
- A core reference text for credible net zero action to bring clarity to what good looks like
- Draws on the 33 most prominent net zero initiatives and frameworks in the existing net zero governance ecosystem to harmonise best practice in one place
 - ISSB is a disclosure framework
 - SBTi is a target-setting framework
 - ISO NZG consolidate both aspects of net zero action, and more
- **Now being converted into the world's first international, independently verifiable net zero standard**, aimed to support organizations of all sizes, sectors and jurisdictions



What will the ISO Net Zero Standard cover?

The international standard will cover the entire journey of net zero action for organisations to support a coherent, net zero aligned approach



Early adoption of the Guidelines

To generate practitioner insights and feedback to inform the future standard's development, we have worked with organisations to become early adopters of the Guidelines on their road to net zero, including:

- General Motors (GM)
- Federation Internationale de l'Automobile (FIA)
- Planet Mark
- Climate Action for Associations (CAFA)

The second cohort of case studies will be published in 2025.

[BSI published a full case study report at London Climate Action Week 2024.](#)



"The ISO Net Zero Guidelines can enhance our decarbonization strategy, and show us that with collaboration, the path ahead is getting clearer"

Kristen Siemen
Chief Sustainability Officer
General Motors



"Our CFO said it was the most self-explanatory strategy he's ever seen and that's because I was following the Guidelines" - Head of Sustainability, second cohort case study organization, December 2024

The ISO Net Zero Standard development process is currently taking place!



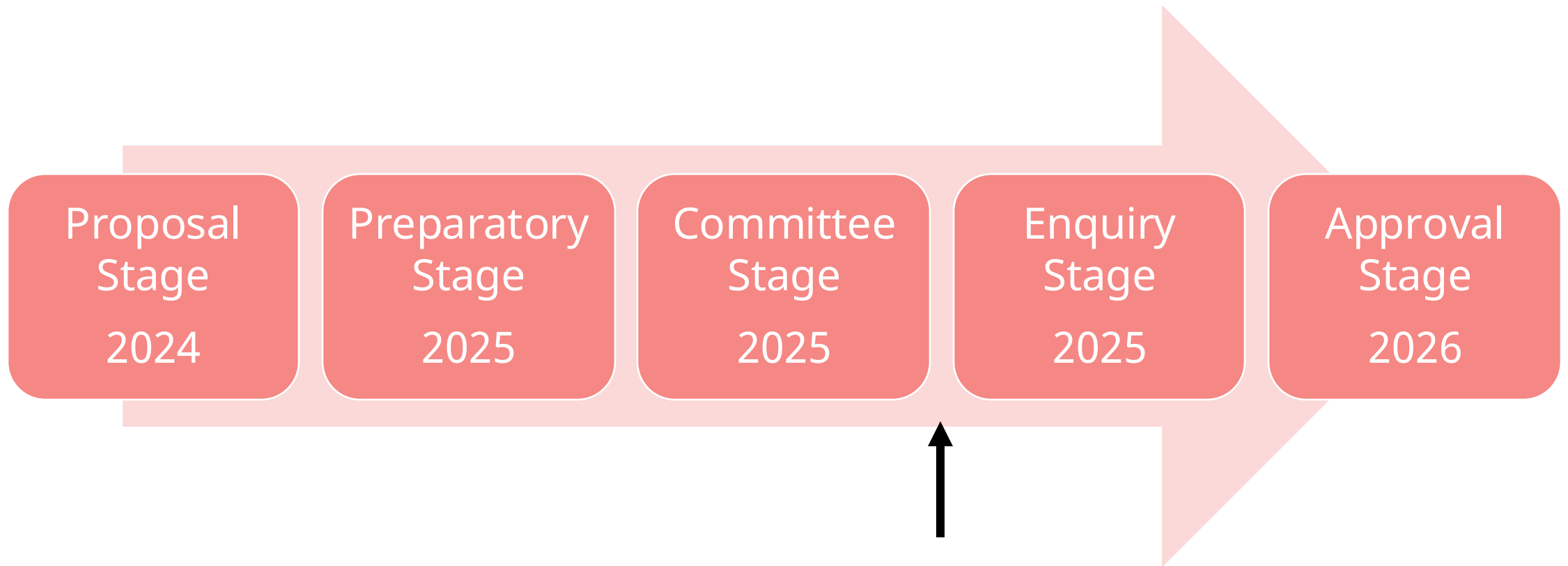
Published on 27 June 2024

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Work is underway to develop ISO's first international standard on net zero. It will provide clarity on the net zero transition, robust requirements, and ultimately enable independently verified and comprehensive climate action and accelerate progress towards a sustainable world.

- Publicly launched at London Climate Action Week in 2024
- The development process is being co-led by BSI, the UK's National Standards Body, and ICONTEC, Colombia's National Standards Body
- This process will convert the ISO Net Zero Guidelines into an independently verifiable ISO standard
- Experts and organizations are encouraged to participate!
- Contact your National Standards Body and refer to committee ISO/TC207/SC7 and Project ISO 14060

What are the timelines of this international process?



What differentiates ISO standards from other voluntary “standards”?

Global scaling potential based on ISO member structure (170+ countries)

Global Quality Infrastructure: system of standards, accreditation and assurance bodies will support trust, accountability and recognition

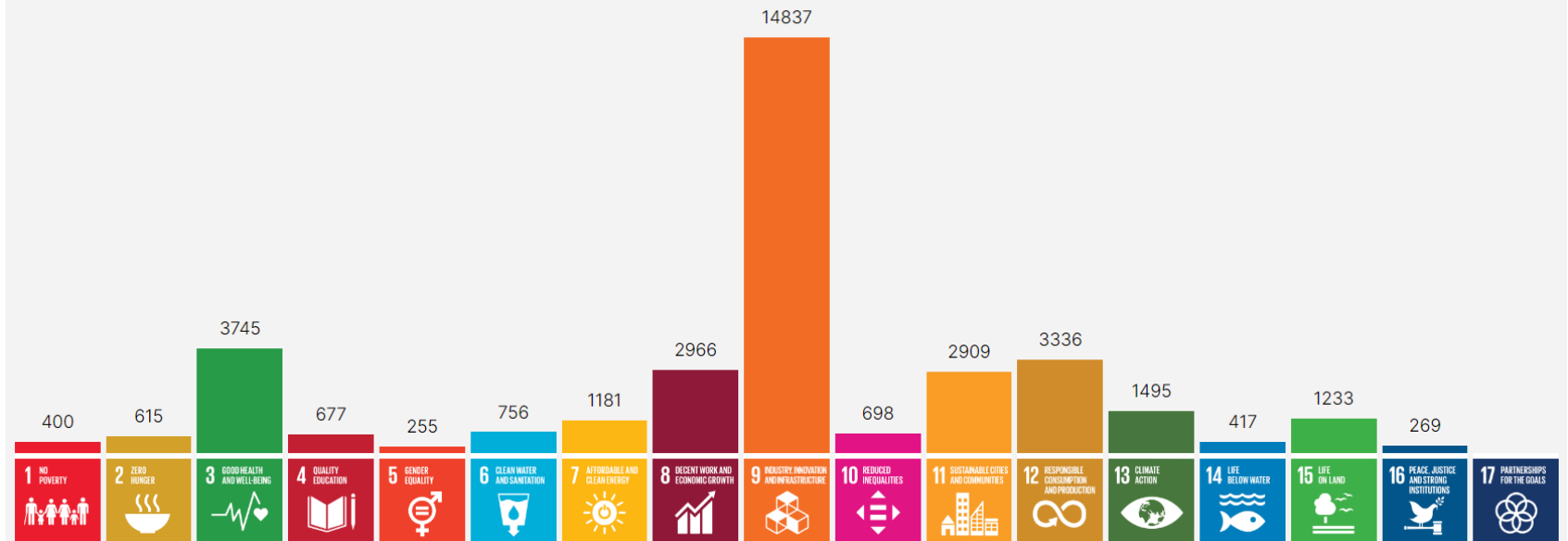
Recognized international standards support interoperability and economic cooperation in emerging net zero policy and regulation (i.e. WTO Technical Barriers to Trade mechanism)

When things don't work as they should, it often means that standards are absent.

”

ISO contributes to all of the Sustainable development goals

Here you can see the number of ISO standards that apply to each Goal.



Conclusion

- The ISO Net Zero Guidelines are being converted into the world's first verifiable international standard on net zero for organizations
 - Leveraging the international standards system will provide much-needed trust, scaling potential, and accountability to net zero governance
 - **Individuals and organizations are encouraged to participate! Contact your National Standards Body and refer to committee ISO/TC207/SC7 and Project ISO 14060**
- The purpose of the international standard system is to converge, harmonize and scale best practice in a way that is fit for purpose for industry actors
- The international standard may have enormous implications for the market



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